Victor H. Zendejas Lopez

Numerical PDE Methods | GPU Scientific Computing | Computational Fluid Dynamics

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EDUCATION

California Institute of Technology, Pasadena, CA — Ph.D., Mechanical Engineering, expected 2027
M.S., Mechanical Engineering, 2023
Coursework: ACM/EE 106ab (Numerical Linear Algebra & PDEs); CS 179 (GPU Programming, CUDA); CS 130 (Software Engineering, Git)

University of California, Berkeley, Berkeley, CA — B.S., Mechanical Engineering, 2020

Chabot College, Hayward, CA — A.S., Mathematics, 2017

PROFILE

I develop PDE-based simulation software combining finite-element methods, GPU programming, and large-scale parallel computing. My CFD research uses MFEM, C++/Python, and CUDA/RAJA, with an emphasis on convergence verification and performance analysis.

TECHNICAL SKILLS

Programming & HPC: C++, Python, CUDA, RAJA, MPI, Fortran, Linux, Git, CMake

Numerical methods: Numerical linear algebra, continuous and discontinuous Galerkin (CG/DG) finite elements (MFEM), ALE hydrodynamics, DNS, LES

Scientific software & tools: MFEM, cuBLAS, cuFFT, MARBL, Caliper, heFFTe, AMReX, PeleLM, VisIt, ParaView

SELECTED GPU SOLVER PROJECT

2D Incompressible Navier–Stokes Solver | CUDA C++ (graduate project)

- I implemented GPU-based three-stage Runge–Kutta time stepping and pressure projection, using **cuBLAS** for stage updates and **cuFFT** for a Fourier-based pressure Poisson solve.
- I wrote CUDA kernels for velocity divergence, spectral Poisson operations, and pressure-gradient correction, managing device memory and integrating kernels with GPU numerical libraries.

RESEARCH & SCIENTIFIC SOFTWARE EXPERIENCE

California Institute of Technology — 2024–Present
Ph.D. Researcher | Numerical PDE methods, turbulence, and HPC

- I develop MFEM-based incompressible Navier–Stokes simulation workflows, assessing discretization effects, numerical dissipation, and turbulence statistics.
- I develop **RAJA GPU kernels** for turbulence statistics and use **Caliper** to profile routines and kernels, measure execution times, and check performance improvements.
- I implemented C++ turbulence diagnostics in a research branch of LLNL's **MARBL** code, using finite-element velocity gradients and MPI reductions to measure higher-order statistics, dissipation, and pressure work on moving meshes.
- I performed mesh-refinement and convergence studies of turbulence quantities and spatial fields (including Helmholtz decomposition), and checked analytical tests, serial/MPI agreement, and CPU/GPU consistency.
- I analyzed simulations up to 1024^3 **effective resolution** using **5,760 MPI ranks on 60 nodes** and developed distributed Python/heFFTe tools for FFTs, energy spectra, and velocity-gradient invariants.

RESEARCH & SCIENTIFIC SOFTWARE EXPERIENCE *(continued)*

Lawrence Livermore National Laboratory (LLNL) 2022–2026
Research Intern | Strategic Deterrence | Four appointments

- I developed two planar Richtmyer–Meshkov benchmarks (single shock and reshock) for MARBL; both configurations were merged into the main codebase and are used by researchers studying shock-driven instabilities.
- I evaluated sensitivity to artificial-viscosity parameters through mesh-refinement studies and cross-code comparisons, matching reference late-stage mixing-layer growth in a 2D single-shock case and comparing 3D flow behavior qualitatively.
- I implemented diagnostics for compressible turbulence and shock visualization in supersonic Taylor–Green vortex studies.

Lawrence Berkeley National Laboratory (LBNL) 2019–2021
Computational Science Researcher

- I developed parallel particle-force calculations and visualization tools in C++ and AMReX for simulations of cancer-cell migration.
- I analyzed hydrogen-enriched flame behavior with PeleLM direct numerical simulations, comparing transport models and interpreting changes in flame structure.

University of California, Berkeley 2018–2020
Undergraduate Researcher | Computational mechanics

- I simulated buckling in architected microlattices using ANSYS and contributed results to a Bayesian design-optimization study published in *Science Advances*.

General Motors — Chassis Engineer (2019): I characterized nonlinear suspension-component response experimentally.

Intel — Autonomous Vehicle Engineer (2018): I prototyped vehicle components and prepared engineering documentation.

PUBLICATIONS & MANUSCRIPTS

V. Z. Lopez, K. Beckwith, R. Rieben, and D. Meiron. "Implicit large-eddy simulation of decaying turbulence: The low-Mach Taylor–Green vortex instability." *Manuscript in preparation.*

Vangelatos, Z., H. M. Sheikh, P. S. Marcus, C. P. Grigoropoulos, **V. Z. Lopez**, et al. "Strength through defects: A novel Bayesian approach for the optimization of architected materials." *Science Advances* 7(41), eabk2218 (2021).

SELECTED HONORS

DOE Computational Science Graduate Fellowship 2021–2025

Caltech Engineering and Applied Science Chair Scholars Fellowship 2021–Present

GEM Associate Fellowship 2021